

Master Thesis-

Topic: Template-Aware Handwritten Text Recognition for Structured Forms

Problem: Recognition of handwritten entries in predefined fields of fixed-layout bank/government-style forms.

Fall Semester: Literature → baselines → forms → preliminary research contribution.

Spring Semester: Main experiments → evaluation → paper/thesis → demo → defense.

Current progress: TrOCR/Colab environment established and initial inference experiments performed.

Fall 2026 — Foundation, Baselines & Research Direction

August–September: Literature + foundations

- Review handwritten OCR/HTR literature.
- Study CNN, CRNN/LSTM + CTC, Transformers and TrOCR.
- Study IAM and relevant form/document datasets.
- Learn OpenCV preprocessing and document alignment.
- Establish evaluation metrics: CER, WER, field-level accuracy.
- Maintain literature review and experimental documentation.
- Identify limitations reported in existing HTR/form-recognition literature.
- Maintain a list of candidate research gaps and hypotheses.

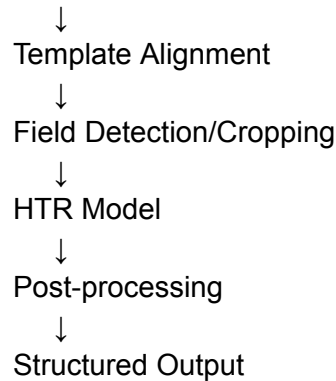
September–October: Baseline implementation

- Build reproducible TrOCR baseline.
- Experiment with IAM or another established handwriting dataset.
- Implement/test CRNN as a traditional deep-learning baseline.
- Compare CRNN vs TrOCR.
- Investigate preprocessing effects on real handwriting.
- Perform systematic error analysis of baseline models.
- Identify failure cases specific to structured forms.
- Formulate 2–3 candidate hypotheses based on observed limitations.

October–November: Structured forms

- Develop a fixed-template form pipeline:

Scanned Form



- Decide on appropriate dataset strategy.
- Begin collecting/creating structured handwritten form samples if needed.
- Test different field types: names, dates, numbers, addresses.
- Evaluate whether different field types exhibit different recognition-error patterns.
- Select the most promising hypothesis.
- Design controlled experiments to test it.

November–December: Hypothesis Validation + Preliminary Research Contribution

- Identify the specific research gap based on literature and baseline results.
- Select one focused improvement, potentially:
 - template-aware recognition,
 - field-aware recognition,
 - preprocessing/noise robustness,
 - domain adaptation/fine-tuning.
- Run preliminary experiments.
- Document results.
- Prepare research progress report/presentation.
- If results are promising, start an IEEE-style paper outline.

Spring 2027 — Main Research, Evaluation & Thesis

January–February: Main implementation

- Finalize dataset and experimental protocol.
- Implement proposed approach.
- Fine-tune/train models.
- Run controlled experiments.
- Maintain reproducibility of experiments.

February–March: Evaluation

- Compare against baselines.
- Evaluate CER/WER and field-level accuracy.
- Perform error analysis.
- Run ablation experiments where appropriate.
- Test robustness to handwriting variation, scan quality, skew/noise, etc.

March–April: Thesis + publication

- Finalize results.
- Write methodology and experimental chapters.
- Prepare conference/IEEE manuscript if contribution/results justify publication.
- Advisor review and revisions.

April–May: Completion

- Complete thesis.
- Final experiments if requested.
- Prepare demo.
- Prepare defense presentation.
- Thesis defense and final submission.

Questions:

Scope and expectations

- Does this scope seem appropriate for a master's thesis?

Timeline

- What milestones would you like me to reach by the end of this semester?
- What should I ideally have completed before winter break?
- When should I start writing the actual thesis?
- Are there departmental proposal, committee, defense, or graduation deadlines I should plan around?

Meetings

- How frequently would you prefer that we meet—weekly, biweekly, or based on milestones?
- What would you like me to bring to our meetings: experiment results, written progress reports, GitHub updates, papers/questions?

Research/publication

- Do you think this project has potential for a conference paper?
- Should publication be one of our objectives, or should I first focus entirely on establishing the research contribution?
- Are there conferences or workshops you would recommend?
- Are there submission deadlines we should design the experiments around?

Dataset

- Should I primarily use established datasets such as IAM initially?
- For the structured-form component, would creating our own controlled dataset be appropriate?
- Are there privacy/IRB considerations if volunteers provide handwriting samples?
- Does the university already have relevant datasets/resources?

Compute

- Does the department have GPU resources?
- Is Colab sufficient initially?
- Are there research servers or computing credits available?

Final deliverables

- Do you expect a working software demo in addition to the thesis?
- What does the thesis defense typically look like?
- Would you like the code/dataset to be publicly released?